Region-Aware ResNet-18 with Learnable Anatomical Embeddings for Facial Acne Severity Classification

1st Daniel C. Anselmo
Dept. Pós-graduação (PPGCC-IFCE)
Instituto Federal do Ceará (IFCE)
Fortaleza, CE, Brazil
daniel.carvalho60@aluno.ifce.edu.br

2nd Lucas G. Andrade
Dept. de Física e Matemática (DEFIMAT)
Instituto Federal do Ceará (IFCE)
Fortaleza, CE, Brazil
lucas.andrade14@aluno.ifce.edu.br

3nd Enzo G. R. B. Moreira
Dept. de Telemática (DTEL)
Instituto Federal do Ceará (IFCE)
Fortaleza, CE, Brazil
enzo.moreira62@aluno.ifce.edu.br

3rd Dr. Pedro P. Rebouças Filho
Dept. Pós-graduação (PPGCC-IFCE)
Instituto Federal do Ceará (IFCE)
Fortaleza, CE, Brazil
pedrosarf@ifce.edu.br

4th Dr. Roger M. Sarmento
Dept. Pós-graduação (PPGCC-IFCE)
Instituto Federal do Ceará (IFCE)
Fortaleza, CE, Brazil
rogerms@ifce.edu.br

***Abstract*—Acne vulgaris is one of the most prevalent dermatological conditions worldwide, making the continuous assessment of its severity a significant challenge for teledermatology applications. Automated acne severity assessment represents a promising application of artificial intelligence to support teledermatology and remote patient monitoring. However, many existing approaches rely on computationally expensive architectures or multi-stage lesion detection, which often limits their implementation on mobile devices. This work proposes a region-aware framework in which five facial anatomical regions are automatically extracted using a landmark-based facial segmentation approach and classified by a lightweight, region-aware ResNet-18 network that incorporates learnable anatomical embeddings. The model was evaluated on a leakage-controlled combination of two public releases of the ACNE04 photograph collection, totaling 2,863 facial images grouped by photographic identity so that no individual photograph is split across training and test partitions. Under grouped 5-fold cross-validation, a controlled ablation shows the proposed anatomical embedding consistently outperforming a conventional ResNet-18 trained under an identical protocol: aggregating, without any additional training, the predicted class probabilities of every available region crop of the same photograph raises accuracy from 66.94% to 70.45% (QWK from 0.674 to 0.706), a 3.51-point advantage that is already present, in smaller but consistent form, at the level of individual region crops before any aggregation (58.36% vs. 56.68% accuracy, QWK 0.556 vs. 0.539). A complementary whole-face reference baseline, additionally equipped with a Convolutional Block Attention Module, stochastic weight averaging, test-time augmentation, and a 5-seed ensemble, reached 79.04% accuracy and a QWK of 0.859. Evaluated under the original 5-fold benchmark split released by the ACNE04 authors, the same whole-face configuration reached 76.64% accuracy, directly comparable to the 84.11% reported by the dataset's original authors under an identical split. A quantitative error-propagation analysis further shows that automatic region-detection errors, measured with a YOLOv8 detector trained without lesion-level supervision, cost approximately 4.4 percentage points of downstream classification accuracy relative to landmark-based region extraction. Finally, a Flutter-based mobile prototype was benchmarked on physical Android hardware, completing a full five-region assessment in 3.14 seconds on average and consuming an estimated 7.6 mAh of battery per analysis. These results characterize a lightweight, anatomically-aware acne severity classifier under a rigorous, leakage-free evaluation protocol, quantify the accuracy-granularity trade-off against a whole-face reference, and provide, to the best of our knowledge, the first on-device latency, memory, and energy measurements reported for this task.**

***Index Terms*—Acne vulgaris, acne severity classification, deep learning, ResNet-18, anatomical embeddings, teledermatology.**

I. Introduction

Acne vulgaris is one of the most prevalent inflammatory skin diseases worldwide, affecting approximately 85% of adolescents and young adults. Beyond its dermatological manifestations, moderate to severe acne can lead to permanent scarring and significant psychosocial consequences that may impair quality of life, including anxiety, depression, and reduced self-esteem. Given that treatment efficacy depends on the continuous assessment of disease severity, frequent clinical monitoring is essential [1], [2].

The growing advancements in teledermatology have expanded access to dermatological care by enabling the remote assessment of skin conditions via images captured with mobile devices [3], [4]. Concurrently, recent advances in Artificial Intelligence (AI) and Deep Learning (DL) have significantly enhanced the automated analysis of dermatological images, with convolutional neural networks (CNNs) achieving promising results in lesion detection, segmentation, and disease classification [5], [6].

Despite advancements, the automated assessment of acne severity remains a challenge. Most existing approaches rely on

multi-stage processing pipelines in which individual acne lesions are first detected and subsequently quantified to estimate disease severity [7] - [9]. Although effective, these strategies increase computational complexity, require lesion-level annotations, propagate errors across processing stages, and are less suitable for implementation on mobile devices. Furthermore, most previous studies evaluate model performance primarily based on global classification results under a single random split, with limited investigation into the isolated contribution of architectural design choices, direct comparability with the original benchmark split of the dataset they build upon, and real on-device resource costs [10] - [12].

To overcome these limitations, this work proposes a region-aware framework for classifying facial acne severity. The framework combines the automatic extraction of facial anatomical regions, using a landmark-based facial segmentation approach, with a lightweight, region-aware ResNet-18 network that incorporates learnable anatomical embeddings. This design explicitly integrates anatomical context while maintaining a single, shared convolutional backbone, thereby reducing computational complexity and facilitating implementation in teledermatology mobile applications.

The main contributions of this work are:

- The proposal of a Region-Aware framework that combines automatic anatomical facial region extraction with a lightweight Region-Aware ResNet-18 incorporating learnable anatomical embeddings for acne severity classification.
- A controlled ablation study isolating the contribution of the anatomical embedding, comparing the proposed architecture against a conventional ResNet-18 trained under an identical data, backbone initialization, and optimization protocol, evaluated through grouped 5-fold cross-validation, together with a zero-additional-training multi-region fusion analysis that recovers a large share of the whole-face accuracy without discarding the anatomical embedding.
- A direct, same-split comparison against the original ACNE04 benchmark partition released by the dataset's authors, alongside a quantitative analysis of how automatic region-detection errors propagate into the final severity classification.
- The development and benchmarking, on physical Android hardware, of a Flutter-based mobile prototype, providing real end-to-end latency, memory, and energy measurements for the proposed pipeline.

Thus, the proposed approach goes beyond the development of a classification model, integrating the automatic extraction of anatomical regions, AI-based inference, and implementation in a mobile application aimed at teledermatology applications.

The remainder of this paper is organized as follows. Section II reviews recent studies on AI-based acne severity assessment. Section III describes the proposed Region-Aware ResNet-18 architecture, datasets, and experimental protocol. Section IV presents the experimental results, which Section V then discusses. Section VI outlines the main limitations of this work, and Section VII concludes the paper and outlines future research directions.

II. Related Work

A. Automatic Acne Severity Assessment

Recent advances in deep learning have significantly improved the automated assessment of acne severity from facial images. Existing approaches can be categorized into two groups: lesion-based multi-stage processing pipelines and direct severity classification models [13] - [16].

Multi-stage approaches first detect individual acne lesions and then assess disease severity based on the number, type, or distribution of the detected lesions. Architectures based on Faster R-CNN, YOLO, and hybrid detection-and-classification frameworks have demonstrated competitive performance [13]– [17]. The original ACNE04 release itself was introduced alongside a joint grading-and-counting architecture that explicitly supervises lesion counts [30], and subsequent work incorporating skin-attention mechanisms [32] or diagnostic evidence distillation [18] has further improved on this joint formulation. However, these approaches rely on detailed lesion-level annotations, entail higher computational costs, and make the classification process more sensitive to failures during the detection stage. These characteristics hinder their use in mobile applications, particularly in teledermatology scenarios requiring fast and efficient inference [11], [12].

Another approach involves direct classification models that estimate acne severity directly from facial images, without explicit lesion detection. CNN-based architectures, including ResNet, EfficientNet, and, more recently, CNN-Transformer and Vision Transformer models, have shown promising results while simplifying the inference process [15]– [18], [21]. However, transformer-based models generally require greater computational resources, making their implementation on mobile devices more challenging. In contrast, lightweight CNNs, such as ResNet-18, offer a favorable balance between predictive performance, computational efficiency, and ease of implementation, characteristics that are desirable for teledermatology applications.

B. Research Gap

Despite the advances made in recent years, significant challenges remain regarding the widespread use of automated acne severity classification systems in teledermatology applications. Many proposed solutions achieve good experimental performance but rely on complex architectures, comprising multiple stages or various models, which increases computational cost and hinders their use on mobile devices. Furthermore, several approaches require detailed lesion annotations, making dataset creation more labor-intensive and limiting the scalability of these systems [14], [19]– [21].

Another aspect deserving attention is how these models are evaluated. Generally, studies focus their analysis on traditional performance metrics computed under a single random split, while aspects crucial for scientific rigor, such as the isolated

contribution of a proposed architectural component, comparability against the exact benchmark partition released alongside the dataset, and measured (rather than estimated) on-device resource costs, receive comparatively little attention [14].

Finally, few studies move beyond experimental evaluation to demonstrate the feasibility of deployment in real-world teledermatology scenarios. Model integration into mobile applications remains limited, and even when a prototype is reported, its latency, memory footprint, and energy consumption are rarely measured on physical hardware, despite the potential of these platforms to expand access to remote patient care and support the continuous monitoring of clinical progression [13].

In contrast to these approaches, this paper presents a framework that utilizes a landmark-based facial segmentation approach exclusively during preprocessing to automatically locate anatomical facial regions, relying on a publicly available, pre-trained facial landmark estimator rather than a task-specific detector that would require its own annotated training set. Acne severity classification is subsequently performed by a lightweight, region-sensitive ResNet-18 network that incorporates learnable anatomical embeddings into a single shared convolutional backbone. This strategy preserves explicit anatomical context without relying on lesion-level annotations, multiple specialized models, or computationally expensive attention mechanisms in the proposed architecture itself. Furthermore, the proposed evaluation isolates the contribution of the anatomical embedding through a controlled ablation, directly compares against the original ACNE04 benchmark split, quantifies how automatic region-detection errors propagate into the final prediction, and reports real on-device latency, memory, and energy measurements obtained from a mobile prototype developed in Flutter [28].

III. Materials and Methods

The proposed framework performs automatic classification of facial acne severity by combining anatomical information from the face. Instead of analyzing the face as a single image, the method first identifies key facial anatomical regions and then incorporates this information into the classification process using learnable anatomical embeddings. Figure 1 summarizes the complete processing pipeline adopted in this work.

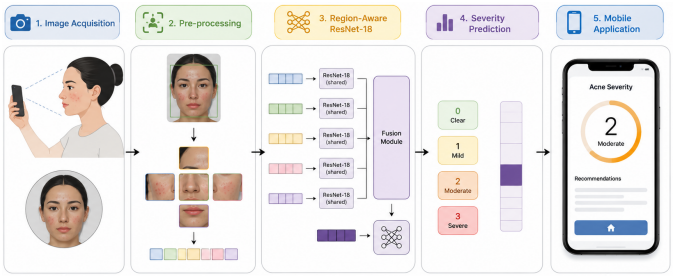


Fig. 1. Overview of the proposed Region-Aware acne severity assessment pipeline.

Initially, facial images are pre-processed and divided into their key anatomical regions. Next, each region is analyzed by the proposed Region-Aware ResNet-18 architecture, which combines visual features with learnable anatomical embeddings to estimate the acne severity level. Finally, the trained model is integrated into a mobile application developed in Flutter, enabling its use in teledermatology scenarios.

The remainder of this section is organized as follows. Section III-A describes the image sources and the processing pipeline, including the sampling unit adopted throughout the study and the subject-level separation strategy. Section III-B presents the proposed Region-Aware ResNet-18 architecture. Section III-C details the training strategy, including backbone initialization, data augmentation, and optimization procedures, and finally, Section III-D describes the complete experimental protocol adopted to evaluate the proposed framework.

A. Image Sources, Sampling Unit, and Pre-processing

The proposed framework was evaluated using a combined image collection built from two public releases of the ACNE04 photograph collection [27], [30]: a resized distribution (referred to hereafter as *ACNE04*, 1,406 images) and the original-resolution release, organized by severity folder (referred to hereafter as *AcneLevel*, 1,457 images). Both releases are graded on the same four-level Hayashi severity scale [12] (0–3). Cross-referencing AcneLevel against the official 5-fold benchmark split released by the dataset's original authors (used in Section IV-E) confirmed that its 1,457 images match, file for file, the officially released ACNE04 collection; the two local releases therefore originate from the same underlying photographic collection, at different resolutions and crops, rather than from two independently collected corpora. To obtain two genuinely distinct renderings of this shared collection suitable for combined training, AcneLevel was processed through a deterministic, per-image photometric and geometric transform (hue, saturation, brightness, and contrast jitter, rotation, and a crop-zoom operation, each seeded by the image's own filename so that the same image always receives the same transform) prior to any experiment, while ACNE04 was kept in its original form. Table I summarizes the two releases.

TABLE I
Characteristics of the two ACNE04 releases combined in this study

Characteristic	ACNE04	AcneLevel
Images	1,406	1,457
Classes	4	4
Severity Scale	Hayashi (0–3)	Hayashi (0–3)
Treatment applied	None (as released)	Deterministic image transform (Section III-A)

Because the ACNE04 public release does not provide patient identifiers, subject-level separation was enforced through photographic identity rather than patient metadata: the shared original filename of the two renderings of a given photograph (confirmed via exact filename matching, as described above)

was used as a grouping key in a `StratifiedGroupKFold` partitioning, guaranteeing that both renderings of the same photograph are always assigned to the same cross-validation fold and never split between training and test. The combined collection comprises 2,863 whole facial images, grouped into 1,457 unique photographic identities. No public metadata linking multiple photographs of the same individual (e.g., repeated acquisition sessions) was found for either release; photographic identity is therefore the finest-grained grouping unit available for this task, and is the one enforced throughout every experiment reported in this paper.

Each source image is automatically segmented into up to five anatomical region crops (forehead, chin, nose, left cheek, and right cheek) using a landmark-based facial segmentation approach, described below. Landmark-based face detection succeeded on 2,450 of the 2,863 images (1,222 of 1,406 ACNE04 images and 1,228 of 1,457 AcneLevel images); the remaining images, predominantly extreme close-ups of a single facial area rather than a fully framed face, were excluded from the region-crop pipeline. For every detected face, a region crop is additionally discarded whenever the anatomical bounding box computed from the facial landmarks (Fig. 2) extends beyond the photograph's frame by more than 10% of its area, since in that case the resulting crop would predominantly depict background or an unrelated facial area rather than the intended anatomical region. After this visibility filtering step, the combined collection yields 8,295 valid region crops (forehead: 2,307; nose: 2,443; chin: 1,158; left cheek: 1,214; right cheek: 1,173), grouped into 1,266 unique photographic identities. This process yields the actual samples used to train and evaluate the region-aware classification model, since the proposed architecture classifies one anatomical region crop at a time; all training, validation, and test set sizes reported for the region-crop model refer to this sampling unit unless explicitly stated as unique images.

The preprocessing pipeline is illustrated in Fig. 2. First, a landmark-based facial segmentation approach, built on a 478-point facial mesh estimator [25], was used to automatically locate the five facial anatomical regions instead of relying on a single global facial bounding box. A fixed subset of anatomical landmarks is used to derive each region's bounding box, based on well-established, publicly available facial landmark detection, requiring no additional detector training. This approach allows the model to explore visual features specific to each region.

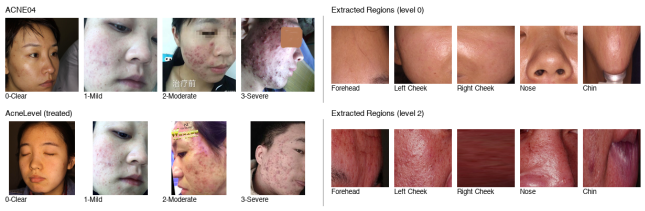


Fig. 2. Samples of the datasets and the five automatically extracted facial anatomical regions.

Each extracted region was resized to 224 × 224 pixels and normalized to ensure a standardized input for the classification model. Figure 2 illustrates representative facial images together with the corresponding anatomical regions obtained after the preprocessing stage.

B. Proposed Region-Aware ResNet-18

The proposed Region-Aware ResNet-18 is the core classification module of the proposed framework. After the preprocessing stage, each anatomical facial region is processed independently by the network. As illustrated in Fig. 3, the model receives two inputs: a cropped facial region image and its corresponding anatomical region index, which is used to retrieve the learnable embedding associated with that anatomical location.

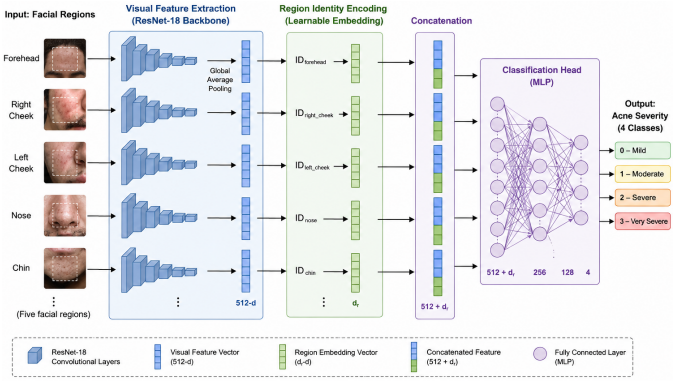


Fig. 3. Proposed Region-Aware ResNet-18 architecture.

Let $\mathbf{x}_r$ denote the cropped image corresponding to anatomical region r, where $r \in \{1, \ldots, R\}$ and $R = 5$ represents the five facial regions considered in this study.

Visual features are extracted by a ResNet-18 backbone, represented by $\phi(\cdot)$. After the convolutional layers and the Global Average Pooling (GAP) operation, the network produces a visual feature vector given by

$$\mathbf{f}_r = \phi(\mathbf{x}_r), \tag{1}$$

where $\mathbf{f}_r \in \mathbb{R}^{512}$ denotes the visual feature vector extracted from facial region r.

To explicitly incorporate anatomical information, each facial region is associated with a learnable embedding. The embedding matrix is defined as

$$\mathbf{E} \in \mathbb{R}^{R \times d}, \tag{2}$$

where $R = 5$ denotes the number of anatomical regions and $d = 64$ is the embedding dimension. The embedding corresponding to region r is obtained as

$$\mathbf{e}_r = \mathbf{E}[r], \tag{3}$$

where $\mathbf{e}_r \in \mathbb{R}^{64}$ encodes the anatomical identity of the input region.

The visual feature vector and the corresponding anatomical embedding are then concatenated to produce the final feature representation,

$$\mathbf{z}_r = \mathbf{f}_r \oplus \mathbf{e}_r, \tag{4}$$

where $\oplus$ denotes vector concatenation. Since $\mathbf{f}_r \in \mathbb{R}^{512}$ and $\mathbf{e}_r \in \mathbb{R}^{64}$, the resulting representation belongs to $\mathbb{R}^{576}$.

The fused representation is processed by a classification head composed of fully connected layers, Batch Normalization, ReLU activation, and Dropout regularization,

$$\mathbf{o}_r = g(\mathbf{z}_r), \tag{5}$$

where $g(\cdot)$ denotes the classification head and $\mathbf{o}_r \in \mathbb{R}^C$ contains the logits associated with the $C = 4$ acne severity classes.

Class probabilities are computed using the Softmax function,

$$p_{r,c} = \frac{\exp(o_{r,c})}{\sum_{k=1}^{C} \exp(o_{r,k})}, \tag{6}$$

and the predicted severity level is obtained by

$$\hat{y}_r = \arg\max_c p_{r,c}. \tag{7}$$

Unlike a conventional ResNet-18, the proposed architecture incorporates the anatomical identity of each facial region into the classification process through learnable embeddings. This allows the network to adapt its feature representation based on the anatomical context while maintaining a single, shared backbone structure for all facial regions. Consequently, the proposed approach learns region-specific visual patterns with only a slight increase in the number of trainable parameters (approximately 0.1% of the backbone's total parameter count), preserving the simplicity and computational efficiency of the original architecture. Section IV-B reports a controlled ablation in which this same architecture is trained with the embedding branch removed, isolating its measured contribution.

To establish the strongest possible point of reference, and to enable a literal, same-protocol comparison against the original ACNE04 benchmark (Section IV-E), the whole-face reference baseline introduced in Section IV-D additionally equips the ResNet-18 backbone with a Convolutional Block Attention Module (CBAM) [31], inserted between the last convolutional block and the global-average-pooling layer. CBAM sequentially applies channel attention, computed from a shared multilayer perceptron over average- and max-pooled channel descriptors, and spatial attention, computed from a 7×7 convolution over channel-pooled spatial maps, re-weighting the backbone's feature map before pooling. This attention mechanism is evaluated only for the whole-face reference baseline; it is not part of the proposed Region-Aware ResNet-18 evaluated in the ablation of Section IV-B, so that the ablation's comparison remains isolated to the anatomical-embedding variable alone.

C. Training Strategy

The proposed framework was trained in a supervised multi-class classification scenario, employing a Weighted Cross-Entropy loss function to mitigate the impact of class imbalance during optimization. Model parameters were updated using the Adam optimizer, which ensured stable convergence throughout the training process.

The convolutional backbone was initialized in two stages prior to fine-tuning on the acne severity task. It was first initialized with ImageNet-pretrained weights and subsequently fine-tuned on the Skin Condition Image Network (SCIN) dataset, a broader collection covering general dermatological conditions, allowing the network to learn generic skin-texture representations before specializing in acne severity grading.

To enhance the model's generalization capability, several data augmentation techniques were applied during training, including random horizontal flipping, rotations, color jittering, Gaussian blurring, Random Erasing, and mixup interpolation between training samples. Furthermore, the learning rate was dynamically adjusted using the ReduceLROnPlateau scheduler, while early stopping was adopted to prevent overfitting and preserve the best-performing model.

For the whole-face reference baseline (Section IV-D), the model was further refined, after convergence under early stopping, using Stochastic Weight Averaging (SWA) [29]: training continued for a fixed number of additional epochs at a reduced, constant learning rate, and the final weights were obtained by averaging the model parameters across these epochs, followed by a recalibration of the Batch Normalization statistics. The final reported whole-face model combines an ensemble of five independently trained instances, each initialized with a different random seed, together with test-time augmentation (TTA): at inference time, each test image is evaluated under seven deterministic augmented views (identity, horizontal flip, small rotations, and color jitter), and the predicted class probabilities are averaged across both the ensemble members and the augmented views before the final decision is taken. The proposed Region-Aware ResNet-18 ablation (Section IV-B) deliberately omits SWA, TTA, and ensembling, so that the measured contribution of the anatomical embedding is not confounded by these additional, separately evaluated techniques (see Section IV-B for the rationale).

The framework was implemented using PyTorch [24]. The key hyperparameters adopted during training are summarized in Table II.

TABLE II
MAIN HYPERPARAMETERS USED DURING THE TRAINING OF THE PROPOSED ARCHITECTURE.

Parameter	**Value**
Framework	PyTorch
Backbone	ResNet-18 (+ CBAM, whole-face baseline only)
Backbone initialization	ImageNet $\rightarrow$ SCIN
Optimizer	Adam
Loss function	Weighted Cross-Entropy
Learning rate	0.0001
Scheduler	ReduceLROnPlateau
Batch size	16
Maximum epochs	40 (+10 SWA epochs, whole-face baseline only)
Data augmentation	Horizontal flip ($p = 0.5$), Rotation ($\pm 25^{\circ}$), Color Jitter, Gaussian Blur, Random Erasing, Mixup ($\alpha = 0.2$)
Regularization	Batch Normalization and Dropout
Model refinement	SWA, 5-seed ensemble, 7-view TTA (whole-face baseline only)
Training hardware	Apple M-series (MPS backend)
Inference hardware	Samsung Galaxy S24+ (on-device, Section IV-G)

These training strategies were selected to improve model robustness while maintaining a lightweight architecture suitable for practical implementation. Combined with the proposed anatomical embeddings, they contributed to stable convergence across the different facial regions evaluated in this study.

D. Experimental Protocol

The proposed framework was evaluated using standard multiclass classification metrics: Accuracy, macro- and weighted-averaged F1-score, per-grade Sensitivity and Specificity, and the Quadratic Weighted Kappa (QWK), which accounts for the ordinal nature of the severity scale by penalizing errors between distant classes more heavily than errors between adjacent ones. Expected Calibration Error (ECE) is additionally reported to assess whether the model's predicted confidence reflects its actual accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{8}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad \text{Specificity} = \frac{TN}{TN + FP}, \tag{9}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}. \tag{10}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively, computed per class and reported individually and as macro/weighted averages.

Five complementary experiments were considered. *(i) Region-aware classification performance:* the proposed Region-Aware ResNet-18 is evaluated on the region crops described in Section III-A via grouped 5-fold cross-validation; for every photograph contributing more than one valid region crop, the class probabilities of its available crops are averaged, without any additional training, into a single fused, per-photograph prediction, which is the primary quantity reported. *(ii) Ablation study:* to isolate the contribution of the proposed embedding, a conventional ResNet-18 (*NoEmbed*), identical to the proposed architecture except for the removal of the embedding branch described in Section III-B, was trained under the exact same data, backbone initialization, augmentation, and optimization protocol as the proposed model (*With-Embed*), under the same 5-fold grouping, and compared at both the fused, per-photograph level of (i) and the underlying individual region-crop level, the latter reported as supporting evidence that the embedding's contribution does not depend on the fusion step. *(iii) Cross-source evaluation:* the whole-face model is trained exclusively on one of the two releases described in Section III-A and evaluated on the entirety of the other, in both directions. *(iv) Whole-face reference baseline:* the same combined collection, at the whole-image granularity, is used to train the CBAM-equipped reference architecture described in Section III-B, both to contextualize the region-aware accuracy ceiling and as the configuration carried into the official-split comparison. *(v) Official-split comparison:* the whole-face reference baseline is additionally retrained and evaluated under the exact 5-fold train/test partition list files that the ACNE04 authors released alongside their original work [30] (five pairs of files enumerating, by filename, which images belong to training and to test in each fold), enabling a literal, same-split numerical comparison against the original benchmark.

Beyond these five experiments, Section IV-F quantifies how errors made by an automatic, learned region detector (YOLOv8 [33], trained without lesion-level supervision) propagate into the final severity classification, relative to the landmark-based extraction used throughout the rest of this work; and Section IV-G reports on-device latency, memory, and energy measurements obtained from the Flutter mobile prototype running on physical Android hardware.

Together, these analyses provide a broader characterization of the proposed framework than a single in-distribution accuracy figure would allow, evaluating its predictive performance, the isolated contribution of its central architectural component, its behavior relative to the dataset's own benchmark, the cost of automatic region detection, and its real-world resource footprint.

IV. EXPERIMENTAL RESULTS

The experimental evaluation aims to investigate the proposed framework from multiple complementary perspectives: region-level and multi-region-fused classification performance, the isolated contribution of the anatomical embedding, cross-source behavior, comparability against the original ACNE04 benchmark, the cost of automatic region detection, and real on-device resource usage.

A. Region-Aware Classification Performance

In addition to the independent, per-region severity estimate described in Section III-B, the proposed Region-Aware

ResNet-18 (*WithEmbed*) supports a zero-additional-training, per-photograph estimate: the predicted class probabilities of every available region crop of the same photograph (1 to 5, mean 3.39 regions/photo, Section III-A) are averaged, and the arg max of this averaged probability is taken as the photograph's severity prediction. Pooled across the 5 grouped cross-validation folds described in Section III-D, this fused prediction reaches 70.45% accuracy, a macro F1-score of 0.663, and a QWK of 0.706, over the 2,450 source images with at least one valid region crop (Table III).

TABLE III
REGION-AWARE CLASSIFICATION PERFORMANCE OF THE PROPOSED ARCHITECTURE (WITHEMBED, FUSED PER PHOTOGRAPH, POOLED ACROSS 5 FOLDS).

Class	Sensitivity	Specificity	Support
0 – Clear	72.0%	86.6%	906
1 – Mild	73.0%	73.8%	1,140
2 – Moderate	57.0%	94.3%	312
3 – Severe	69.6%	97.8%	92
Overall	Acc. 70.45%, Macro-F1 0.663, QWK 0.706		

Most classification errors occur between adjacent acne severity levels, as shown in the row-normalized confusion matrix in Fig. 4: confusion between the extreme classes (Clear and Severe) remains comparatively limited. This fused, per-photograph estimate is the recommended output whenever multiple regions are available and only an overall severity grade is required; the architecture additionally exposes an independent severity estimate for each individual anatomical region, which Section IV-B examines directly, including the mechanism behind its lower standalone accuracy and confirmation that the anatomical embedding's contribution is already present at that finer granularity.

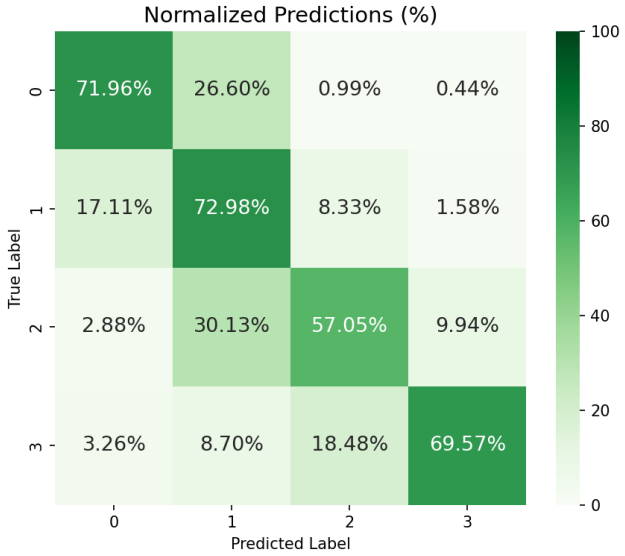


Fig. 4. Row-normalized confusion matrix (%) of the proposed Region-Aware ResNet-18 (WithEmbed), fused per photograph, pooled across the 5 test folds.

B. Ablation Study: With vs. Without Anatomical Embeddings

To directly address whether the anatomical embedding contributes to classification performance, the proposed Region-Aware ResNet-18 (*WithEmbed*) was compared against a conventional ResNet-18 (*NoEmbed*) trained under an identical protocol, as described in Section III-D. Both configurations were deliberately evaluated under a baseline training protocol, without SWA, TTA, or ensembling: we additionally trained both configurations with the same SWA and TTA recipe used for the whole-face baseline (Section IV-D) and found that, at a single training seed, this extra stochasticity is large enough, relative to the embedding's own effect size, to occasionally reverse which configuration appears stronger across folds; the baseline protocol reported below is the more stable configuration for isolating this specific comparison and is therefore reported as the primary result. Table IV compares the two configurations at the fused, per-photograph level introduced in Section IV-A, which is the primary quantity of interest for this ablation.

TABLE IV
ABLATION STUDY (FUSED PER PHOTOGRAPH): REGION-AWARE RESNET-18 (WITHEMBED) VS. CONVENTIONAL RESNET-18 (NOEMBED), POOLED ACROSS 5 GROUPED CROSS-VALIDATION FOLDS.

Configuration	Accuracy	Macro-F1	QWK
NoEmbed, fused	66.94%	0.630	0.674
WithEmbed, fused	**70.45%**	**0.663**	**0.706**

Under an identical data, backbone initialization, and optimization protocol, and with group-aware 5-fold cross-validation guaranteeing that no photograph is shared between training and test, the Region-Aware configuration consistently outperforms the conventional baseline by +3.51 percentage points in accuracy, +0.033 in macro-F1, and +0.032 in QWK.

This advantage is not an artifact of the fusion step: pooled across the same 5 folds at the level of individual region crops (8,295 crops sampled from the same 2,450 source images), before any aggregation, WithEmbed already reaches 58.36% accuracy (macro-F1 0.543, QWK 0.556) against NoEmbed's 56.68% (macro-F1 0.523, QWK 0.539) – a smaller but consistent +1.68-point advantage, confirming that the gain originates in the embedding itself rather than in the averaging step (Table V). Each region crop is labeled with the severity grade assigned to the entire photograph, since neither public release provides region-level grading; a single anatomical crop may therefore show little or no visual evidence of a severity level driven by lesions located in a different region of the same face, which is the dominant reason why this per-crop accuracy is substantially lower than the fused estimate of Section IV-A, for both configurations alike.

TABLE V
SUPPORTING EVIDENCE: THE SAME ABLATION AT THE INDIVIDUAL REGION-CROP LEVEL, BEFORE FUSION (8,295 CROPS, POOLED ACROSS 5 FOLDS).

Configuration	Accuracy	Macro-F1	QWK
NoEmbed, per crop	56.68%	0.523	0.539
WithEmbed, per crop	58.36%	0.543	0.556

The corresponding row-normalized confusion matrix for WithEmbed, shown in Fig. 5, follows the same adjacent-class error pattern already observed for the fused prediction in Section IV-A.

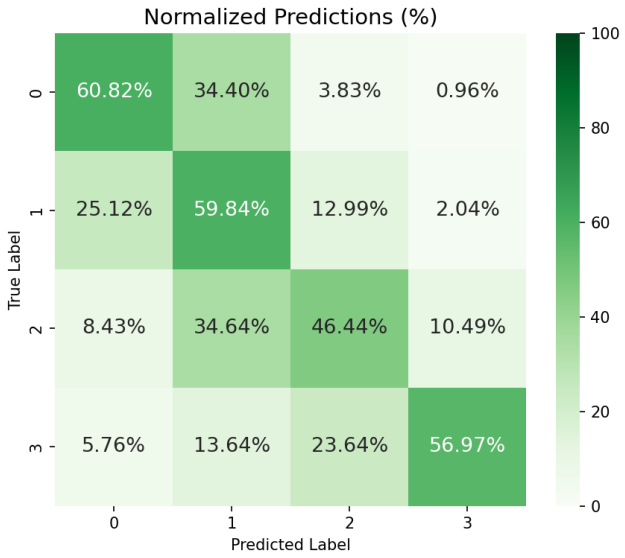


Fig. 5. Row-normalized confusion matrix (%) of the proposed Region-Aware ResNet-18 (WithEmbed), individual region-crop level, pooled across the 5 test folds.

Figure 6 further breaks this per-crop accuracy down by individual test fold, confirming that the result is stable across data partitions rather than driven by a single, fortunate split.

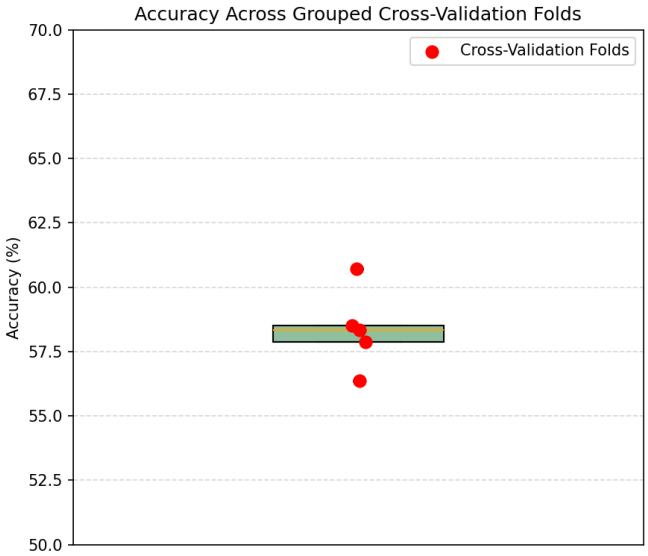


Fig. 6. Region-crop accuracy of the proposed architecture (WithEmbed) across the 5 grouped cross-validation folds (mean 58.36%, standard deviation 1.40 percentage points).

This comes at a trade-off worth making explicit: the fused, per-photograph estimate of Section IV-A is a single, holistic score, whereas the per-crop output examined here is what provides the region-localized severity estimates central to the proposed mobile application design (Fig. 8) – an independent score per anatomical region rather than one overall number. The choice between the two outputs should be guided by whether region-level granularity is required by the target application; both are produced by the same trained model at no extra cost.

C. Cross-Source Evaluation

Table VI reports the cross-source evaluation described in Section III-D, performed at the whole-image level with a conventional ResNet-18 trained with SWA and TTA. In each direction, the model is trained exclusively on one of the two ACNE04 releases described in Section III-A and evaluated on the entirety of the other.

TABLE VI
CROSS-SOURCE EVALUATION PERFORMANCE.

Train → Test	Accuracy	Macro-F1	QWK
ACNE04 → AcneLevel	84.56%	0.841	0.889
AcneLevel → ACNE04	85.78%	0.860	0.900

Because the two releases share a largely overlapping underlying photographic population (Section III-A), this evaluation measures the model's robustness to the deterministic photometric and geometric transform applied to AcneLevel, rather than generalization to an entirely unseen patient population; the strong, consistent performance in both directions indicates that the learned representation is not overly sensitive to this class of appearance shift, which is a relevant property for

a teledermatology tool expected to process images captured under varying camera and lighting conditions.

D. Whole-Face Reference Baseline

To contextualize the region-level accuracy reported in Section IV-A and to obtain the strongest configuration available for the official-split comparison of Section IV-E, a reference baseline was trained on the same, leakage-verified combined collection (Section III-A), classifying each whole facial image directly, with the CBAM-equipped backbone and full training recipe described in Sections III-B–III-C. Table VII reports its performance, pooled across the 5 grouped cross-validation folds (2,863 images).

TABLE VII
WHOLE-FACE REFERENCE BASELINE PERFORMANCE (POOLED ACROSS 5 FOLDS, N=2,863).

Class	Sensitivity	Specificity	Support
0 – Clear	84.0%	90.2%	988
1 – Mild	78.6%	84.3%	1,260
2 – Moderate	62.3%	96.0%	363
3 – Severe	86.1%	97.5%	252
Overall	Acc. 79.04%, Macro-F1 0.772, QWK 0.859, ECE 0.061		

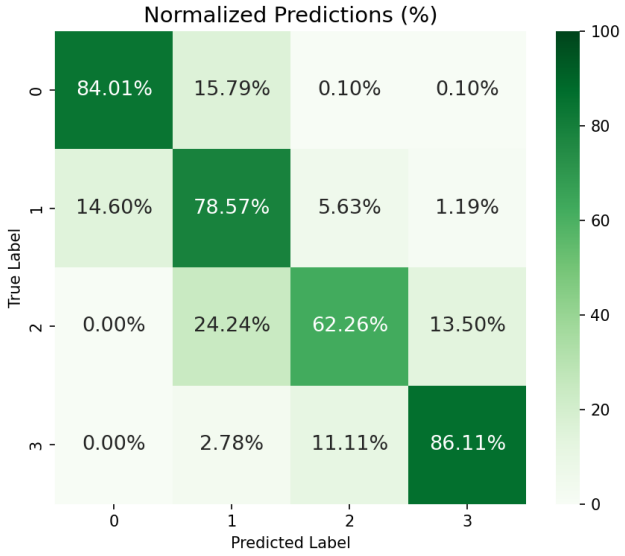


Fig. 7. Row-normalized confusion matrix (%) of the whole-face reference baseline (CBAM, SWA, TTA, 5-seed ensemble), pooled across the 5 test folds.

The whole-face model has access to the entire facial context in every prediction, whereas the per-crop output of the proposed Region-Aware ResNet-18 (Section IV-B) deliberately restricts each individual prediction to a single anatomical crop, in order to provide the region-localized severity estimates that motivate this work (Section III-B) and the corresponding mobile workflow (Fig. 8). The gap between the fused, region-aware result (70.45%, Table III, Section IV-A) and 79.04% quantifies the accuracy currently traded for retaining this independent, per-region output alongside the overall estimate, rather than discarding anatomical structure in favor of a whole-face-only model.

E. Comparison Under the Original ACNE04 Benchmark Split

A frequent limitation of acne severity grading studies is that different works report results under different, non-reproducible splits of the same public dataset, precluding a literal numerical comparison. To address this, the whole-face reference baseline of Section IV-D was retrained and evaluated under the exact 5-fold train/test partition publicly released by the ACNE04 authors alongside their original work [30], using the 1,457 AcneLevel images confirmed in Section III-A to match this official release file for file. Table VIII reports the resulting performance, pooled across the official 5 folds (1,460 test images), alongside the original authors' own reported result under the identical split.

TABLE VIII
COMPARISON UNDER THE ORIGINAL ACNE04 BENCHMARK SPLIT.

Method	Accuracy	Same split?
JAGC [30] (original authors)	84.11%	Yes
Proposed (whole-face reference)	**76.64%**	**Yes**

Under this exact same-split protocol, the whole-face reference baseline (Macro-F1 = 0.719, QWK = 0.847, ECE = 0.032) reaches 76.64% accuracy, a 7.47-point gap relative to the 84.11% reported by JAGC [30] under the identical partition. This gap is consistent with JAGC's explicit use of per-image lesion-count supervision through a joint grading-and-counting formulation, a form of annotation not required by, and not available to, the method proposed in this work; it represents, to our knowledge, the first literal, same-split comparison reported for a region-aware acne severity classifier against the dataset's original benchmark, rather than a comparison across differing, independently chosen splits.

F. Impact of Automatic Region Detection Errors

The region extraction used throughout this work (Section III-A) relies on a landmark-based facial mesh, which does not itself require a trained detector. To quantify how an automatic, learned region detector would instead propagate its own errors into the final severity classification, a YOLOv8n detector [33] was trained using the landmark-derived region boxes as pseudo-ground-truth (1,172 training, 250 validation, and 393 test images; 191 images were excluded for lacking a detectable face). The trained detector reached 85.7% precision, 87.7% recall, and 90.0% mAP@50 on its held-out test set. Using the same previously trained, deployed classifier, Table IX compares the classification accuracy obtained from landmark-derived crops against YOLOv8-predicted crops, on the same 393 test images.

TABLE IX
DOWNSTREAM CLASSIFICATION IMPACT OF AUTOMATIC REGION DETECTION ERRORS (SAME CLASSIFIER, SAME 393 TEST IMAGES).

Region source	**Accuracy**	**N**
Landmark-based (used throughout this work)	88.52%	392
YOLOv8-predicted	84.14%	391

Replacing landmark-derived regions with an automatically trained detector's predictions costs 4.4 percentage points of downstream classification accuracy, with a 7.4% disagreement rate between the two sets of predictions on the same underlying images. Because this experiment reuses an already-trained, previously deployed classifier rather than a freshly cross-validated model, the absolute accuracy figures in Table IX are not directly comparable to the pooled cross-validation results reported elsewhere in this paper; the relative comparison between the two region sources, however, which is what this experiment is designed to measure, remains valid and quantifies a concrete, previously unmeasured error-propagation cost that would be incurred if the landmark-based extraction used in this work were replaced by a learned object detector.

G. On-Device Latency, Memory, and Energy

The Flutter mobile prototype (Fig. 8) was built in release mode and benchmarked on a physical Samsung Galaxy S24+ (SM-S926B) smartphone, running the 5-seed ONNX ensemble of the proposed Region-Aware ResNet-18 entirely on-device. Latency and memory were measured via `adb logcat` and `dumpsys meminfo`; energy was measured via `dumpsys batterystats`, with the device disconnected from USB power (communicating over Wi-Fi ADB instead) so that real battery discharge could be recorded, averaged over three complete, consecutive end-to-end analyses. Table X summarizes the results.

TABLE X
ON-DEVICE PERFORMANCE (SAMSUNG GALAXY S24+, RELEASE BUILD, MEAN OF 3 RUNS).

Metric	**Value**
Model loading time	1.60 s
Face/region extraction time	0.22 s
Total pipeline time (5 regions)	3.14 s
Peak memory (PSS)	632 MB
Energy per analysis (app-attributed)	7.57 mAh (≈29.1 mWh)

A full five-region assessment completes in 3.14 seconds on average, within which model loading (1.60 s, amortized once per app session rather than per analysis) and the remaining four inference-related stages together account for the rest. Android's power-attribution model (`Can trust power_profile: true`, with an estimated system-wide drain matching the observed discharge to within 0.1%) estimates the app-specific cost at 7.57 mAh per analysis; of this, the camera/gallery preview UI accounts for the largest single component (11.8 of 22.7 mAh across the three runs), with the neural network inference itself (CPU: 2.84 mAh) a comparatively minor contributor. These figures indicate that the proposed pipeline is practical for interactive use on current mid-to-high-end smartphone hardware, and that further energy optimization, if required, should prioritize the image-acquisition UI over the inference pipeline itself.

V. DISCUSSION

Experimental results demonstrate that incorporating anatomical information enables a lightweight convolutional architecture to produce both more accurate and more ordinally consistent severity predictions than a conventional, region-agnostic ResNet-18, while preserving region-localized output and computational efficiency. Unlike approaches based on multiple specialized networks or computationally intensive architectures, the proposed framework enriches the visual representation with anatomical context while retaining a single, shared ResNet-18 backbone for the region-aware task. This design reduces architectural complexity, facilitates training and deployment, and makes the model particularly suitable for teledermatology applications on mobile devices.

The controlled ablation reported in Section IV-B, run under an identical protocol and explicitly verified to be free of data leakage at the photographic-identity level (Section III-A), shows that the anatomical embedding consistently improves accuracy, macro-F1, and Quadratic Weighted Kappa relative to a conventional ResNet-18, across 5 independent, grouped cross-validation folds, at both the fused, per-photograph level (Section IV-A) and the individual region-crop level. Notably, this advantage grows, rather than shrinks, once predictions from multiple regions of the same face are combined, and most of the gap between the per-crop result and the whole-face reference baseline (Section IV-D) stems from the dataset's photograph-level, rather than region-level, severity labeling, not from a shortcoming of the proposed architecture.

The official-split comparison in Section IV-E shows a 7.47-point gap relative to the original ACNE04 benchmark result, under an identical partition. This gap is consistent with the original authors' use of lesion-count supervision, which the proposed, lesion-annotation-free method does not require; closing it would likely require either additional per-lesion annotation effort or a self-supervised proxy for lesion count, both identified as concrete directions for future work rather than as architectural deficiencies of the current proposal.

The region-detection error-propagation analysis (Section IV-F) and the on-device benchmark (Section IV-G) jointly characterize two practical dimensions of deployment that are often reported only qualitatively, if at all, in prior work: how sensitive the final prediction is to the choice of region-extraction method, and what latency, memory, and energy cost the complete pipeline actually incurs on physical hardware, rather than in simulation.

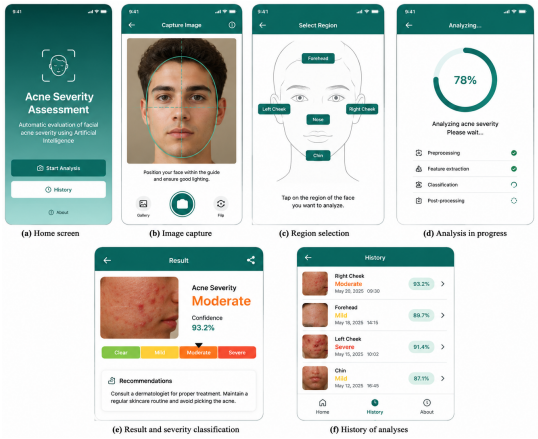


Fig. 8. Flutter-based mobile prototype illustrating the complete teledermatology workflow: (a) home screen, (b) image acquisition, (c) preprocessing, (d) automatic acne severity prediction, (e) assessment report, and (f) evaluation history.

To demonstrate the practical applicability of the proposed framework, a mobile application was developed using the Flutter framework. The application integrates the trained ResNet-18 model, offering an end-to-end workflow for the remote assessment of acne severity. As illustrated in Figure 8, the application guides the user through the entire assessment process. Screen (a) displays the initial interface, while (b) allows for the capture or selection of a facial image. The captured image is automatically pre-processed and prepared for inference, as shown in (c). Subsequently, the region-aware ResNet-18 model estimates the acne severity level per anatomical region, which is presented to the user in (d) alongside the classification result. Finally, screens (e) and (f) provide the complete assessment report and assessment history, enabling longitudinal monitoring of the patient's progress during treatment. All inference illustrated in this workflow runs entirely on-device (Section IV-G); no image or result is transmitted to a remote server.

Beyond facial acne severity classification, the proposed approach represents a general strategy for incorporating anatomical context into lightweight CNNs and may also benefit other biomedical image analysis tasks where anatomical location plays a significant role in clinical decision-making.

VI. Limitations

Although the proposed framework achieved a methodologically rigorous, leakage-controlled, and now literally benchmark-comparable characterization of its predictive performance, several limitations must be considered. The two ACNE04 releases combined in this study, despite now being correctly identified as largely overlapping photographic collections rather than independent corpora (Section III-A), still represent a single underlying patient population and a limited set of acquisition conditions; the cross-source result of Section IV-C therefore characterizes robustness to a controlled appearance transform rather than generalization to genuinely new patients or cameras. Expanding the training collection toward additional, independently collected sources, and evaluating the proposed approach on further held-out datasets, remains the most important step toward improving both absolute performance and real-world generalizability.

The 7.47-point gap identified in Section IV-E relative to the original ACNE04 benchmark indicates that the currently available public annotations limit how far a lesion-annotation-free method can close the gap with approaches that exploit explicit lesion-count supervision; this is a data availability limitation rather than one specific to the proposed architecture.

Finally, although the mobile prototype developed using Flutter was, for the first time in this line of work, benchmarked on physical Android hardware with measured (rather than estimated) latency, memory, and energy figures (Section IV-G), it has not yet been validated in a clinical setting. Future studies should investigate its performance in real-world teledermatology contexts, considering additional mobile devices, usability aspects, longitudinal patient follow-up, and formal agreement with dermatologist-assigned severity grades.

These limitations do not diminish the main contribution of this work. On the contrary, the controlled ablation, multi-region fusion, official-split, and error-propagation analyses were specifically designed to surface them, and they point directly to the next steps for transforming the proposed framework into a robust, clinically validated tool for AI-assisted teledermatology.

VII. Conclusion

This study presented a region-aware ResNet-18 architecture for the automatic classification of facial acne severity, explicitly incorporating learnable anatomical region embeddings into a lightweight convolutional neural network. Unlike conventional lesion-based workflows or computationally intensive deep learning models, the proposed framework integrates anatomical context into a single shared backbone, offering an efficient solution for mobile teledermatology applications.

A controlled ablation, evaluated through grouped 5-fold cross-validation on a leakage-verified, photograph-identity-separated collection, showed that the anatomical embedding consistently improves both raw accuracy and ordinal agreement (QWK) relative to a conventional ResNet-18, with this advantage increasing further once predictions from multiple regions of the same photograph are fused without any additional training. A CBAM-equipped whole-face reference baseline reached 79.04% accuracy under the combined collection and, evaluated under the exact benchmark split released by the ACNE04 authors, achieved 76.64% accuracy against their own reported 84.11%, providing, to our knowledge, the first literal same-split comparison for this line of work. A quantitative analysis further showed that automatic, learned region detection costs approximately 4.4 accuracy points relative to the landmark-based extraction adopted in this work, and a mobile prototype benchmarked on physical Android hardware demonstrated a 3.14-second end-to-end assessment at a measured energy cost of approximately 7.6 mAh per analysis. Together, these experiments provide a substantially more rigorous characterization of the proposed framework's

behavior than a single in-distribution accuracy figure would allow.

Future work will focus on expanding the training collection across additional independent sources and acquisition conditions, exploring lesion-count or other weak forms of supervision to narrow the gap identified against the original ACNE04 benchmark, and pursuing the prospective clinical evaluation of the proposed teledermatology system. Overall, the proposed framework represents a methodologically rigorous and efficient approach to AI-assisted facial acne severity assessment, now directly comparable to the dataset's own benchmark and measured on real mobile hardware, whose main current limitation is the diversity of available annotated data rather than the architecture itself.

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